

ITC Practical 1

July 3, 2026

```
[86]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import sklearn
```

0.1 EXTRACTION OF STRUCTURED DATA INTO A DATAFRAME USING PANDAS

```
[88]: TitanDF=pd.read_csv('tested.csv')
```

0.2 CHECKING IF DATA IS EMPTY OR NOT

```
[89]: TitanDF.empty
```

```
[89]: False
```

```
[90]: TitanDF.head()
```

```
[90]:
```

	PassengerId	Survived	Pclass	\
0	892	0	3	
1	893	1	3	
2	894	0	2	
3	895	0	3	
4	896	1	3	

	Name	Sex	Age	SibSp	Parch	\
0	Kelly, Mr. James	male	34.5	0	0	
1	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	
2	Myles, Mr. Thomas Francis	male	62.0	0	0	
3	Wirz, Mr. Albert	male	27.0	0	0	
4	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	

	Ticket	Fare	Cabin	Embarked
0	330911	7.8292	NaN	Q
1	363272	7.0000	NaN	S
2	240276	9.6875	NaN	Q
3	315154	8.6625	NaN	S
4	3101298	12.2875	NaN	S

```
[91]: TitanDF.tail()
```

```
[91]:
```

	PassengerId	Survived	Pclass	Name	Sex	\
413	1305	0	3	Spector, Mr. Woolf	male	
414	1306	1	1	Oliva y Ocana, Dona. Fermina	female	
415	1307	0	3	Saether, Mr. Simon Sivertsen	male	
416	1308	0	3	Ware, Mr. Frederick	male	
417	1309	0	3	Peter, Master. Michael J	male	

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
413	NaN	0	0	A.5. 3236	8.0500	NaN	S
414	39.0	0	0	PC 17758	108.9000	C105	C
415	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	NaN	0	0	359309	8.0500	NaN	S
417	NaN	1	1	2668	22.3583	NaN	C

```
[92]: TitanDF[180:200]
```

```
[92]:
```

	PassengerId	Survived	Pclass	\
180	1072	0	2	
181	1073	0	1	
182	1074	1	1	
183	1075	0	3	
184	1076	1	1	
185	1077	0	2	
186	1078	1	2	
187	1079	0	3	
188	1080	1	3	
189	1081	0	2	
190	1082	0	2	
191	1083	0	1	
192	1084	0	3	
193	1085	0	2	
194	1086	0	2	
195	1087	0	3	
196	1088	0	1	
197	1089	1	3	
198	1090	0	2	
199	1091	1	3	

	Name	Sex	Age	SibSp	\
180	McCrie, Mr. James Matthew	male	30.0	0	
181	Compton, Mr. Alexander Taylor Jr	male	37.0	1	
182	Marvin, Mrs. Daniel Warner (Mary Graham Carmic...	female	18.0	1	
183	Lane, Mr. Patrick	male	NaN	0	
184	Douglas, Mrs. Frederick Charles (Mary Helene B...	female	27.0	1	
185	Maybery, Mr. Frank Hubert	male	40.0	0	

186	Phillips, Miss. Alice Frances Louisa	female	21.0	0
187	Davies, Mr. Joseph	male	17.0	2
188	Sage, Miss. Ada	female	NaN	8
189	Veal, Mr. James	male	40.0	0
190	Angle, Mr. William A	male	34.0	1
191	Salomon, Mr. Abraham L	male	NaN	0
192	van Billiard, Master. Walter John	male	11.5	1
193	Lingane, Mr. John	male	61.0	0
194	Drew, Master. Marshall Brines	male	8.0	0
195	Karlsson, Mr. Julius Konrad Eugen	male	33.0	0
196	Spedden, Master. Robert Douglas	male	6.0	0
197	Nilsson, Miss. Berta Olivia	female	18.0	0
198	Baimbrigge, Mr. Charles Robert	male	23.0	0
199	Rasmussen, Mrs. (Lena Jacobsen Solvang)	female	NaN	0

	Parch	Ticket	Fare	Cabin	Embarked
180	0	233478	13.0000	NaN	S
181	1	PC 17756	83.1583	E52	C
182	0	113773	53.1000	D30	S
183	0	7935	7.7500	NaN	Q
184	1	PC 17558	247.5208	B58 B60	C
185	0	239059	16.0000	NaN	S
186	1	S.O./P.P. 2	21.0000	NaN	S
187	0	A/4 48873	8.0500	NaN	S
188	2	CA. 2343	69.5500	NaN	S
189	0	28221	13.0000	NaN	S
190	0	226875	26.0000	NaN	S
191	0	111163	26.0000	NaN	S
192	1	A/5. 851	14.5000	NaN	S
193	0	235509	12.3500	NaN	Q
194	2	28220	32.5000	NaN	S
195	0	347465	7.8542	NaN	S
196	2	16966	134.5000	E34	C
197	0	347066	7.7750	NaN	S
198	0	C.A. 31030	10.5000	NaN	S
199	0	65305	8.1125	NaN	S

```
[93]: TitanDF.shape
```

```
[93]: (418, 12)
```

```
[94]: TitanDF.columns
```

```
[94]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
        'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
        dtype='object')
```

```
[95]: TitanDF.dtypes
```

```
[95]: PassengerId      int64
      Survived        int64
      Pclass          int64
      Name            object
      Sex             object
      Age            float64
      SibSp           int64
      Parch           int64
      Ticket          object
      Fare            float64
      Cabin           object
      Embarked        object
      dtype: object
```

```
[96]: TitanDF.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     418 non-null   int64
1   Survived        418 non-null   int64
2   Pclass          418 non-null   int64
3   Name            418 non-null   object
4   Sex             418 non-null   object
5   Age            332 non-null   float64
6   SibSp           418 non-null   int64
7   Parch           418 non-null   int64
8   Ticket          418 non-null   object
9   Fare            417 non-null   float64
10  Cabin           91 non-null    object
11  Embarked        418 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
```

```
[97]: TitanDF.isnull().any()
```

```
[97]: PassengerId      False
      Survived        False
      Pclass          False
      Name            False
      Sex             False
      Age             True
      SibSp           False
      Parch           False
```

```
Ticket      False
Fare        True
Cabin       True
Embarked    False
dtype: bool
```

```
[98]: TitanDF.isna().sum()
```

```
[98]: PassengerId      0
      Survived        0
      Pclass          0
      Name            0
      Sex             0
      Age            86
      SibSp           0
      Parch           0
      Ticket          0
      Fare            1
      Cabin          327
      Embarked        0
      dtype: int64
```

```
[99]: TitanDF.Sex
```

```
[99]: 0      male
      1      female
      2      male
      3      male
      4      female
      ...
      413    male
      414    female
      415    male
      416    male
      417    male
      Name: Sex, Length: 418, dtype: object
```

```
[100]: TitanDF['Age'].mean()
```

```
[100]: np.float64(30.272590361445783)
```

```
[101]: TitanDF['Age'].median()
```

```
[101]: 27.0
```

```
[102]: TitanDF['Fare'].max()
```

```
[102]: 512.3292
```

```
[103]: TitanDF['Fare'].min()
```

```
[103]: 0.0
```

```
[104]: TitanDF.loc [TitanDF['Fare'] ==0]
```

```
[104]:
```

	PassengerId	Survived	Pclass	Name	\
266	1158	0	1	Chisholm, Mr. Roderick Robert Crispin	
372	1264	0	1	Ismay, Mr. Joseph Bruce	

	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
266	male	NaN	0	0	112051	0.0	NaN	S
372	male	49.0	0	0	112058	0.0	B52 B54 B56	S

```
[105]: TitanDF.loc [TitanDF['Fare'] >400]
```

```
[105]:
```

	PassengerId	Survived	Pclass	\
343	1235	1	1	

	Name	Sex	Age	SibSp	\
343	Cardeza, Mrs. James Warburton Martinez (Charlo...	female	58.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
343	1	PC 17755	512.3292	B51 B53 B55	C

```
[106]: TitanDF.loc[(TitanDF['Fare']>=1)&(TitanDF['Fare']<50)]
```

```
[106]:
```

	PassengerId	Survived	Pclass	\
0	892	0	3	
1	893	1	3	
2	894	0	2	
3	895	0	3	
4	896	1	3	
..	...	...	...	
412	1304	1	3	
413	1305	0	3	
415	1307	0	3	
416	1308	0	3	
417	1309	0	3	

	Name	Sex	Age	SibSp	Parch	\
0	Kelly, Mr. James	male	34.5	0	0	
1	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	
2	Myles, Mr. Thomas Francis	male	62.0	0	0	
3	Wirz, Mr. Albert	male	27.0	0	0	
4	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	
..	...	...	...	...	...	
412	Henriksson, Miss. Jenny Lovisa	female	28.0	0	0	

413	Spector, Mr. Woolf	male	NaN	0	0
415	Saether, Mr. Simon Sivertsen	male	38.5	0	0
416	Ware, Mr. Frederick	male	NaN	0	0
417	Peter, Master. Michael J	male	NaN	1	1

	Ticket	Fare	Cabin	Embarked
0	330911	7.8292	NaN	Q
1	363272	7.0000	NaN	S
2	240276	9.6875	NaN	Q
3	315154	8.6625	NaN	S
4	3101298	12.2875	NaN	S
..	...	...	...	...
412	347086	7.7750	NaN	S
413	A.5. 3236	8.0500	NaN	S
415	SOTON/O.Q. 3101262	7.2500	NaN	S
416	359309	8.0500	NaN	S
417	2668	22.3583	NaN	C

[334 rows x 12 columns]

```
[107]: TitanDF.describe()
```

```
[107]:
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	418.000000	418.000000	418.000000	332.000000	418.000000
mean	1100.500000	0.363636	2.265550	30.272590	0.447368
std	120.810458	0.481622	0.841838	14.181209	0.896760
min	892.000000	0.000000	1.000000	0.170000	0.000000
25%	996.250000	0.000000	1.000000	21.000000	0.000000
50%	1100.500000	0.000000	3.000000	27.000000	0.000000
75%	1204.750000	1.000000	3.000000	39.000000	1.000000
max	1309.000000	1.000000	3.000000	76.000000	8.000000

	Parch	Fare
count	418.000000	417.000000
mean	0.392344	35.627188
std	0.981429	55.907576
min	0.000000	0.000000
25%	0.000000	7.895800
50%	0.000000	14.454200
75%	0.000000	31.500000
max	9.000000	512.329200

```
[108]: TitanDF['Pclass'].value_counts()
```

```
[108]: Pclass
3      218
1      107
```

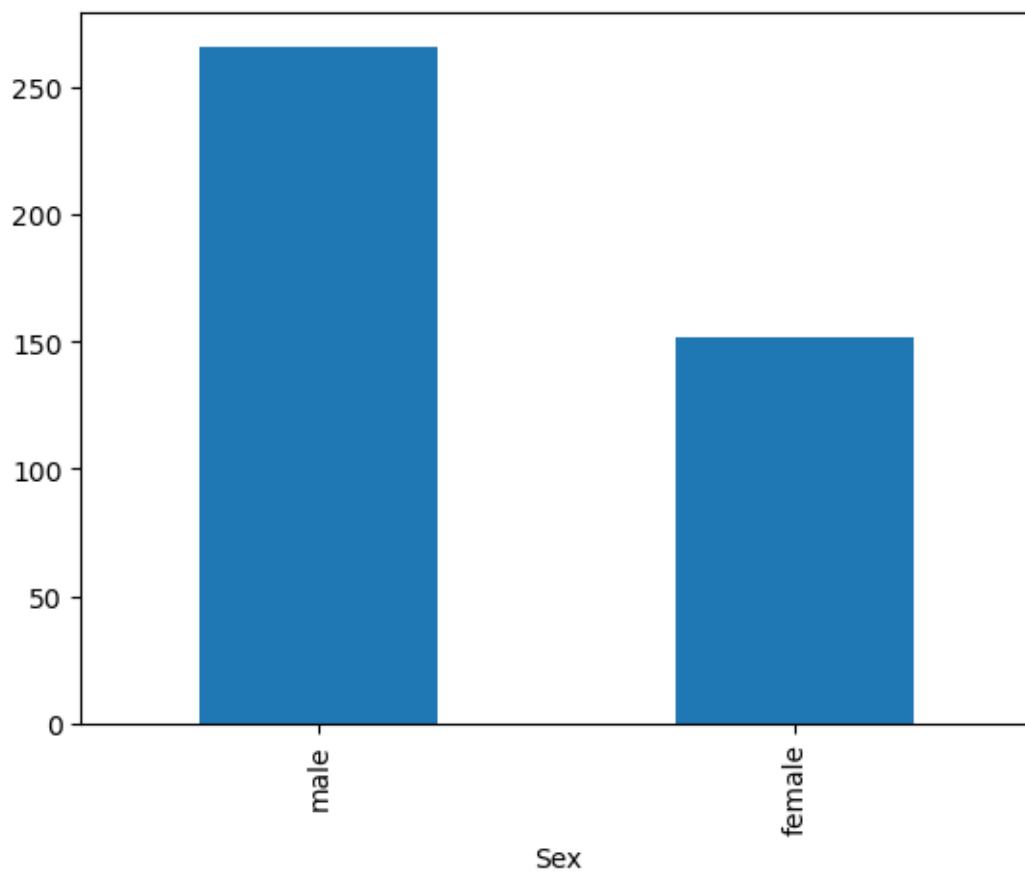
```
2      93
Name: count, dtype: int64
```

```
[109]: TitanDF['Sex'].value_counts()
```

```
[109]: Sex
male      266
female    152
Name: count, dtype: int64
```

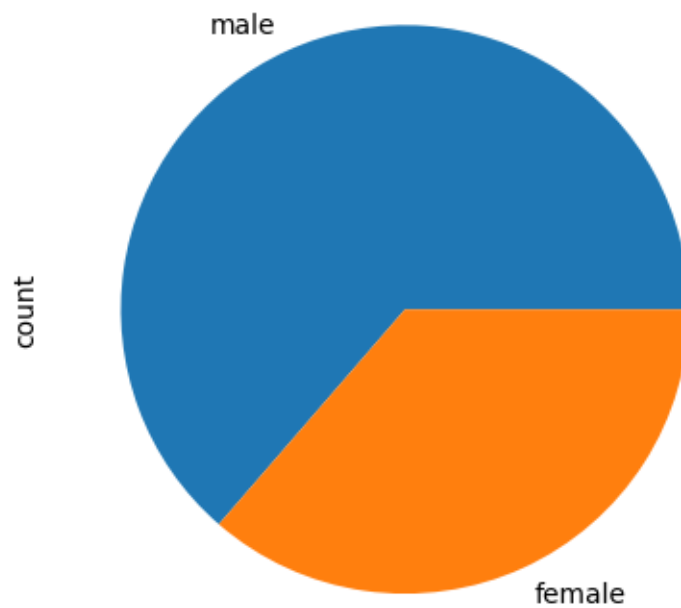
```
[110]: TitanDF['Sex'].value_counts().plot(kind='bar')
```

```
[110]: <Axes: xlabel='Sex'>
```



```
[111]: TitanDF['Sex'].value_counts().plot(kind='pie')
```

```
[111]: <Axes: ylabel='count'>
```

```
[112]: TitanDF['Age'].describe()
```

```
[112]: count      332.000000
      mean       30.272590
      std        14.181209
      min         0.170000
      25%        21.000000
      50%        27.000000
      75%        39.000000
      max        76.000000
      Name: Age, dtype: float64
```

```
[113]: TitanDF['Fare'].describe()
```

```
[113]: count      417.000000
      mean       35.627188
      std        55.907576
      min         0.000000
      25%         7.895800
      50%        14.454200
      75%        31.500000
      max       512.329200
      Name: Fare, dtype: float64
```

```
[114]: TitanDF['Survived'].value_counts()
```

```
[114]: Survived
0      266
1      152
Name: count, dtype: int64
```

```
[115]: TitanDF['Parch'].value_counts()
```

```
[115]: Parch
0      324
1       52
2       33
3        3
4        2
9        2
6        1
5        1
Name: count, dtype: int64
```

```
[116]: TitanDF['Embarked'].value_counts()
```

```
[116]: Embarked
S      270
C      102
Q       46
Name: count, dtype: int64
```

```
[117]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Embarked'] == 'S')]
```

```
[117]:
```

	PassengerId	Survived	Pclass	\
1	893	1	3	
4	896	1	3	
12	904	1	1	
14	906	1	1	
18	910	1	3	
..	...	...	...	
391	1283	1	1	
395	1287	1	1	
400	1292	1	1	
409	1301	1	3	
412	1304	1	3	

	Name	Sex	Age	SibSp	\
1	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	
4	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	
12	Snyder, Mrs. John Pillsbury (Nelle Stevenson)	female	23.0	1	
14	Chaffee, Mrs. Herbert Fuller (Carrie Constance...	female	47.0	1	

18		Ilmakangas, Miss. Ida Livija	female	27.0	1
..		...	...	...	...
391		Lines, Mrs. Ernest H (Elizabeth Lindsey James)	female	51.0	0
395		Smith, Mrs. Lucien Philip (Mary Eloise Hughes)	female	18.0	1
400		Bonnell, Miss. Caroline	female	30.0	0
409		Peacock, Miss. Treasteall	female	3.0	1
412		Henriksson, Miss. Jenny Lovisa	female	28.0	0

	Parch		Ticket	Fare	Cabin	Embarked
1	0		363272	7.0000	NaN	S
4	1		3101298	12.2875	NaN	S
12	0		21228	82.2667	B45	S
14	0	W.E.P.	5734	61.1750	E31	S
18	0	STON/O2.	3101270	7.9250	NaN	S
..	...		...	...	...	...
391	1	PC	17592	39.4000	D28	S
395	0		13695	60.0000	C31	S
400	0		36928	164.8667	C7	S
409	1	SOTON/O.Q.	3101315	13.7750	NaN	S
412	0		347086	7.7750	NaN	S

[88 rows x 12 columns]

```
[118]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Embarked'] == 'C')]
```

```
[118]:
```

	PassengerId	Survived	Pclass	\
8	900	1	3	
15	907	1	2	
19	911	1	3	
24	916	1	1	
26	918	1	1	
48	940	1	1	
59	951	1	1	
74	966	1	1	
100	992	1	1	
104	996	1	3	
112	1004	1	1	
122	1014	1	1	
150	1042	1	1	
168	1060	1	1	
179	1071	1	1	
184	1076	1	1	
208	1100	1	1	
218	1110	1	1	
220	1112	1	2	
224	1116	1	1	
225	1117	1	3	

239	1131	1	1
240	1132	1	1
249	1141	1	3
272	1164	1	1
283	1175	1	3
296	1188	1	2
314	1206	1	1
333	1225	1	3
343	1235	1	1
347	1239	1	3
350	1242	1	1
361	1253	1	2
364	1256	1	1
368	1260	1	1
371	1263	1	1
375	1267	1	1
397	1289	1	1
402	1294	1	1
414	1306	1	1

	Name	Sex	Age	SibSp	\
8	Abraham, Mrs. Joseph (Sophie Halaut Easu)	female	18.0	0	
15	del Carlo, Mrs. Sebastiano (Argenia Genovesi)	female	24.0	1	
19	Assaf Khalil, Mrs. Mariana (Miriam)"	female	45.0	0	
24	Ryerson, Mrs. Arthur Larned (Emily Maria Borie)	female	48.0	1	
26	Ostby, Miss. Helene Ragnhild	female	22.0	0	
48	Bucknell, Mrs. William Robert (Emma Eliza Ward)	female	60.0	0	
59	Chaudanson, Miss. Victorine	female	36.0	0	
74	Geiger, Miss. Amalie	female	35.0	0	
100	Stengel, Mrs. Charles Emil Henry (Annie May Mo...	female	43.0	1	
104	Thomas, Mrs. Alexander (Thamine Thelma)"	female	16.0	1	
112	Evans, Miss. Edith Corse	female	36.0	0	
122	Schabert, Mrs. Paul (Emma Mock)	female	35.0	1	
150	Earnshaw, Mrs. Boulton (Olive Potter)	female	23.0	0	
168	Cassebeer, Mrs. Henry Arthur Jr (Eleanor Genev...	female	NaN	0	
179	Compton, Mrs. Alexander Taylor (Mary Eliza Ing...	female	64.0	0	
184	Douglas, Mrs. Frederick Charles (Mary Helene B...	female	27.0	1	
208	Rosenbaum, Miss. Edith Louise	female	33.0	0	
218	Widener, Mrs. George Dunton (Eleanor Elkins)	female	50.0	1	
220	Duran y More, Miss. Florentina	female	30.0	1	
224	Candee, Mrs. Edward (Helen Churchill Hungerford)	female	53.0	0	
225	Moubarek, Mrs. George (Omine Amenia" Alexander)"	female	NaN	0	
239	Douglas, Mrs. Walter Donald (Mahala Dutton)	female	48.0	1	
240	Lindstrom, Mrs. Carl Johan (Sigrid Posse)	female	55.0	0	
249	Khalil, Mrs. Betros (Zahie Maria" Elias)"	female	NaN	1	
272	Clark, Mrs. Walter Miller (Virginia McDowell)	female	26.0	1	
283	Touma, Miss. Maria Youssef	female	9.0	1	

296		Laroche, Miss. Louise	female	1.0	1
314		White, Mrs. John Stuart (Ella Holmes)	female	55.0	0
333		Nakid, Mrs. Said (Waika Mary" Mowad)"	female	19.0	1
343	Cardeza, Mrs. James Warburton Martinez (Charlo...	female	58.0	0	
347	Whabee, Mrs. George Joseph (Shawneene Abi-Saab)	female	38.0	0	
350	Greenfield, Mrs. Leo David (Blanche Strouse)	female	45.0	0	
361	Mallet, Mrs. Albert (Antoinette Magnin)	female	24.0	1	
364	Harder, Mrs. George Achilles (Dorothy Annan)	female	25.0	1	
368	Gibson, Mrs. Leonard (Pauline C Boeson)	female	45.0	0	
371	Wilson, Miss. Helen Alice	female	31.0	0	
375	Bowen, Miss. Grace Scott	female	45.0	0	
397	Frolicher-Stehli, Mrs. Maxmillian (Margaretha ...	female	48.0	1	
402	Gibson, Miss. Dorothy Winifred	female	22.0	0	
414	Oliva y Ocana, Dona. Fermina	female	39.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
8	0	2657	7.2292	NaN	C
15	0	SC/PARIS 2167	27.7208	NaN	C
19	0	2696	7.2250	NaN	C
24	3	PC 17608	262.3750	B57 B59 B63 B66	C
26	1	113509	61.9792	B36	C
48	0	11813	76.2917	D15	C
59	0	PC 17608	262.3750	B61	C
74	0	113503	211.5000	C130	C
100	0	11778	55.4417	C116	C
104	1	2625	8.5167	NaN	C
112	0	PC 17531	31.6792	A29	C
122	0	13236	57.7500	C28	C
150	1	11767	83.1583	C54	C
168	0	17770	27.7208	NaN	C
179	2	PC 17756	83.1583	E45	C
184	1	PC 17558	247.5208	B58 B60	C
208	0	PC 17613	27.7208	A11	C
218	1	113503	211.5000	C80	C
220	0	SC/PARIS 2148	13.8583	NaN	C
224	0	PC 17606	27.4458	NaN	C
225	2	2661	15.2458	NaN	C
239	0	PC 17761	106.4250	C86	C
240	0	112377	27.7208	NaN	C
249	0	2660	14.4542	NaN	C
272	0	13508	136.7792	C89	C
283	1	2650	15.2458	NaN	C
296	2	SC/Paris 2123	41.5792	NaN	C
314	0	PC 17760	135.6333	C32	C
333	1	2653	15.7417	NaN	C
343	1	PC 17755	512.3292	B51 B53 B55	C
347	0	2688	7.2292	NaN	C

350	1	PC 17759	63.3583	D10 D12	C
361	1	S.C./PARIS 2079	37.0042	NaN	C
364	0	11765	55.4417	E50	C
368	1	112378	59.4000	NaN	C
371	0	16966	134.5000	E39 E41	C
375	0	PC 17608	262.3750	NaN	C
397	1	13567	79.2000	B41	C
402	1	112378	59.4000	NaN	C
414	0	PC 17758	108.9000	C105	C

```
[119]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Embarked'] == 'Q')]
```

```
[119]:
```

	PassengerId	Survived	Pclass	\
6	898	1	3	
63	955	1	3	
66	958	1	3	
70	962	1	3	
79	971	1	3	
86	978	1	3	
88	980	1	3	
111	1003	1	3	
113	1005	1	3	
127	1019	1	3	
160	1052	1	3	
200	1092	1	3	
206	1098	1	3	
216	1108	1	3	
227	1119	1	3	
273	1165	1	3	
282	1174	1	3	
291	1183	1	3	
304	1196	1	3	
313	1205	1	3	
315	1207	1	3	
408	1300	1	3	
410	1302	1	3	
411	1303	1	1	

	Name	Sex	Age	SibSp	\
6	Connolly, Miss. Kate	female	30.0	0	
63	Bradley, Miss. Bridget Delia	female	22.0	0	
66	Burns, Miss. Mary Delia	female	18.0	0	
70	Mulvihill, Miss. Bertha E	female	24.0	0	
79	Doyle, Miss. Elizabeth	female	24.0	0	
86	Barry, Miss. Julia	female	27.0	0	
88	O'Donoghue, Ms. Bridget	female	NaN	0	
111	Shine, Miss. Ellen Natalia	female	NaN	0	

113	Buckley, Miss. Katherine	female	18.5	0
127	McCoy, Miss. Alicia	female	NaN	2
160	Smyth, Miss. Julia	female	NaN	0
200	Murphy, Miss. Nora	female	NaN	0
206	McGowan, Miss. Katherine	female	35.0	0
216	Mahon, Miss. Bridget Delia	female	NaN	0
227	McNeill, Miss. Bridget	female	NaN	0
273	Lennon, Miss. Mary	female	NaN	1
282	Fleming, Miss. Honora	female	NaN	0
291	Daly, Miss. Margaret Marcella Maggie"	female	30.0	0
304	McCarthy, Miss. Catherine Katie"	female	NaN	0
313	Carr, Miss. Jeannie	female	37.0	0
315	Hagardon, Miss. Kate	female	17.0	0
408	Riordan, Miss. Johanna Hannah"	female	NaN	0
410	Naughton, Miss. Hannah	female	NaN	0
411	Minahan, Mrs. William Edward (Lillian E Thorpe)	female	37.0	1

	Parch	Ticket	Fare	Cabin	Embarked
6	0	330972	7.6292	NaN	Q
63	0	334914	7.7250	NaN	Q
66	0	330963	7.8792	NaN	Q
70	0	382653	7.7500	NaN	Q
79	0	368702	7.7500	NaN	Q
86	0	330844	7.8792	NaN	Q
88	0	364856	7.7500	NaN	Q
111	0	330968	7.7792	NaN	Q
113	0	329944	7.2833	NaN	Q
127	0	367226	23.2500	NaN	Q
160	0	335432	7.7333	NaN	Q
200	0	36568	15.5000	NaN	Q
206	0	9232	7.7500	NaN	Q
216	0	330924	7.8792	NaN	Q
227	0	370368	7.7500	NaN	Q
273	0	370371	15.5000	NaN	Q
282	0	364859	7.7500	NaN	Q
291	0	382650	6.9500	NaN	Q
304	0	383123	7.7500	NaN	Q
313	0	368364	7.7500	NaN	Q
315	0	AQ/3. 30631	7.7333	NaN	Q
408	0	334915	7.7208	NaN	Q
410	0	365237	7.7500	NaN	Q
411	0	19928	90.0000	C78	Q

```
[120]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] > 30)]
```

```
[120]: PassengerId  Survived  Pclass  \
1            893         1       3
```

14	906	1	1
19	911	1	3
24	916	1	1
32	924	1	3
44	936	1	1
48	940	1	1
49	941	1	3
59	951	1	1
69	961	1	1
74	966	1	1
77	969	1	1
96	988	1	1
100	992	1	1
112	1004	1	1
114	1006	1	1
122	1014	1	1
141	1033	1	1
153	1045	1	3
178	1070	1	2
179	1071	1	1
206	1098	1	3
208	1100	1	1
213	1105	1	2
214	1106	1	3
218	1110	1	1
224	1116	1	1
239	1131	1	1
240	1132	1	1
241	1133	1	2
305	1197	1	1
309	1201	1	3
313	1205	1	3
314	1206	1	1
324	1216	1	1
330	1222	1	2
343	1235	1	1
347	1239	1	3
349	1241	1	2
350	1242	1	1
356	1248	1	1
362	1254	1	2
368	1260	1	1
371	1263	1	1
374	1266	1	1
375	1267	1	1
391	1283	1	1
397	1289	1	1

411	1303	1	1
414	1306	1	1

	Name	Sex	Age	SibSp	\
1	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	
14	Chaffee, Mrs. Herbert Fuller (Carrie Constance...	female	47.0	1	
19	Assaf Khalil, Mrs. Mariana (Miriam)"	female	45.0	0	
24	Ryerson, Mrs. Arthur Larned (Emily Maria Borie)	female	48.0	1	
32	Dean, Mrs. Bertram (Eva Georgetta Light)	female	33.0	1	
44	Kimball, Mrs. Edwin Nelson Jr (Gertrude Parsons)	female	45.0	1	
48	Bucknell, Mrs. William Robert (Emma Eliza Ward)	female	60.0	0	
49	Coutts, Mrs. William (Winnie Minnie" Treanor)"	female	36.0	0	
59	Chaudanson, Miss. Victorine	female	36.0	0	
69	Fortune, Mrs. Mark (Mary McDougald)	female	60.0	1	
74	Geiger, Miss. Amalie	female	35.0	0	
77	Cornell, Mrs. Robert Clifford (Malvina Helen L...	female	55.0	2	
96	Cavendish, Mrs. Tyrell William (Julia Florence...	female	76.0	1	
100	Stengel, Mrs. Charles Emil Henry (Annie May Mo...	female	43.0	1	
112	Evans, Miss. Edith Corse	female	36.0	0	
114	Straus, Mrs. Isidor (Rosalie Ida Blun)	female	63.0	1	
122	Schabert, Mrs. Paul (Emma Mock)	female	35.0	1	
141	Daniels, Miss. Sarah	female	33.0	0	
153	Klasen, Mrs. (Hulda Kristina Eugenia Lofqvist)	female	36.0	0	
178	Becker, Mrs. Allen Oliver (Nellie E Baumgardner)	female	36.0	0	
179	Compton, Mrs. Alexander Taylor (Mary Eliza Ing...	female	64.0	0	
206	McGowan, Miss. Katherine	female	35.0	0	
208	Rosenbaum, Miss. Edith Louise	female	33.0	0	
213	Howard, Mrs. Benjamin (Ellen Truelove Arman)	female	60.0	1	
214	Andersson, Miss. Ida Augusta Margareta	female	38.0	4	
218	Widener, Mrs. George Dunton (Eleanor Elkins)	female	50.0	1	
224	Candee, Mrs. Edward (Helen Churchill Hungerford)	female	53.0	0	
239	Douglas, Mrs. Walter Donald (Mahala Dutton)	female	48.0	1	
240	Lindstrom, Mrs. Carl Johan (Sigrid Posse)	female	55.0	0	
241	Christy, Mrs. (Alice Frances)	female	45.0	0	
305	Crosby, Mrs. Edward Gifford (Catherine Elizabe...	female	64.0	1	
309	Hansen, Mrs. Claus Peter (Jennie L Howard)	female	45.0	1	
313	Carr, Miss. Jeannie	female	37.0	0	
314	White, Mrs. John Stuart (Ella Holmes)	female	55.0	0	
324	Kreuchen, Miss. Emilie	female	39.0	0	
330	Davies, Mrs. John Morgan (Elizabeth Agnes Mary...	female	48.0	0	
343	Cardeza, Mrs. James Warburton Martinez (Charlo...	female	58.0	0	
347	Whabee, Mrs. George Joseph (Shawneene Abi-Saab)	female	38.0	0	
349	Walcroft, Miss. Nellie	female	31.0	0	
350	Greenfield, Mrs. Leo David (Blanche Strouse)	female	45.0	0	
356	Brown, Mrs. John Murray (Caroline Lane Lamson)	female	59.0	2	
362	Ware, Mrs. John James (Florence Louise Long)	female	31.0	0	
368	Gibson, Mrs. Leonard (Pauline C Boeson)	female	45.0	0	

371		Wilson, Miss. Helen Alice	female	31.0	0
374		Dodge, Mrs. Washington (Ruth Vidaver)	female	54.0	1
375		Bowen, Miss. Grace Scott	female	45.0	0
391		Lines, Mrs. Ernest H (Elizabeth Lindsey James)	female	51.0	0
397		Frolicher-Stehli, Mrs. Maxmillian (Margaretha ...	female	48.0	1
411		Minahan, Mrs. William Edward (Lillian E Thorpe)	female	37.0	1
414		Oliva y Ocana, Dona. Fermina	female	39.0	0

	Parch	Ticket	Fare	Cabin	Embarked
1	0	363272	7.0000	NaN	S
14	0	W.E.P. 5734	61.1750	E31	S
19	0	2696	7.2250	NaN	C
24	3	PC 17608	262.3750	B57 B59 B63 B66	C
32	2	C.A. 2315	20.5750	NaN	S
44	0	11753	52.5542	D19	S
48	0	11813	76.2917	D15	C
49	2	C.A. 37671	15.9000	NaN	S
59	0	PC 17608	262.3750	B61	C
69	4	19950	263.0000	C23 C25 C27	S
74	0	113503	211.5000	C130	C
77	0	11770	25.7000	C101	S
96	0	19877	78.8500	C46	S
100	0	11778	55.4417	C116	C
112	0	PC 17531	31.6792	A29	C
114	0	PC 17483	221.7792	C55 C57	S
122	0	13236	57.7500	C28	C
141	0	113781	151.5500	NaN	S
153	2	350405	12.1833	NaN	S
178	3	230136	39.0000	F4	S
179	2	PC 17756	83.1583	E45	C
206	0	9232	7.7500	NaN	Q
208	0	PC 17613	27.7208	A11	C
213	0	24065	26.0000	NaN	S
214	2	347091	7.7750	NaN	S
218	1	113503	211.5000	C80	C
224	0	PC 17606	27.4458	NaN	C
239	0	PC 17761	106.4250	C86	C
240	0	112377	27.7208	NaN	C
241	2	237789	30.0000	NaN	S
305	1	112901	26.5500	B26	S
309	0	350026	14.1083	NaN	S
313	0	368364	7.7500	NaN	Q
314	0	PC 17760	135.6333	C32	C
324	0	24160	211.3375	NaN	S
330	2	C.A. 33112	36.7500	NaN	S
343	1	PC 17755	512.3292	B51 B53 B55	C
347	0	2688	7.2292	NaN	C

349	0	F.C.C.	13528	21.0000	NaN	S
350	1	PC	17759	63.3583	D10 D12	C
356	0		11769	51.4792	C101	S
362	0	CA	31352	21.0000	NaN	S
368	1		112378	59.4000	NaN	C
371	0		16966	134.5000	E39 E41	C
374	1		33638	81.8583	A34	S
375	0	PC	17608	262.3750	NaN	C
391	1	PC	17592	39.4000	D28	S
397	1		13567	79.2000	B41	C
411	0		19928	90.0000	C78	Q
414	0	PC	17758	108.9000	C105	C

```
[121]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] < 30)]
```

```
[121]:
```

	PassengerId	Survived	Pclass	\
4	896	1	3	
8	900	1	3	
12	904	1	1	
15	907	1	2	
18	910	1	3	
..	...	...	...	
385	1277	1	2	
395	1287	1	1	
402	1294	1	1	
409	1301	1	3	
412	1304	1	3	

	Name	Sex	Age	SibSp	\
4	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	
8	Abraham, Mrs. Joseph (Sophie Halaut Easu)	female	18.0	0	
12	Snyder, Mrs. John Pillsbury (Nelle Stevenson)	female	23.0	1	
15	del Carlo, Mrs. Sebastiano (Argenia Genovesi)	female	24.0	1	
18	Ilmakangas, Miss. Ida Livija	female	27.0	1	
..	...	...	...	...	
385	Herman, Miss. Kate	female	24.0	1	
395	Smith, Mrs. Lucien Philip (Mary Eloise Hughes)	female	18.0	1	
402	Gibson, Miss. Dorothy Winifred	female	22.0	0	
409	Peacock, Miss. Treasteall	female	3.0	1	
412	Henriksson, Miss. Jenny Lovisa	female	28.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
4	1	3101298	12.2875	NaN	S
8	0	2657	7.2292	NaN	C
12	0	21228	82.2667	B45	S
15	0	SC/PARIS 2167	27.7208	NaN	C
18	0	STON/02. 3101270	7.9250	NaN	S

```

..      ...      ...      ...      ...
385      2      220845  65.0000  NaN      S
395      0      13695  60.0000  C31      S
402      1      112378  59.4000  NaN      C
409      1  SOTON/O.Q. 3101315  13.7750  NaN      S
412      0      347086  7.7750  NaN      S

```

[71 rows x 12 columns]

```
[130]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] < 30) &
↳(TitanDF['Sex'] == 'female')]
```

```
[130]: PassengerId  Survived  Pclass  \
4            896         1         3
8            900         1         3
12           904         1         1
15           907         1         2
18           910         1         3
..          ...          ...      ...
385          1277         1         2
395          1287         1         1
402          1294         1         1
409          1301         1         3
412          1304         1         3

```

```

                                     Name      Sex  Age  SibSp  \
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1
8  Abraham, Mrs. Joseph (Sophie Halaut Easu)    female  18.0      0
12 Snyder, Mrs. John Pillsbury (Nelle Stevenson)  female  23.0      1
15 del Carlo, Mrs. Sebastiano (Argenia Genovesi)  female  24.0      1
18      Ilmakangas, Miss. Ida Livija             female  27.0      1
..          ...          ...      ...
385      Herman, Miss. Kate                      female  24.0      1
395 Smith, Mrs. Lucien Philip (Mary Eloise Hughes)  female  18.0      1
402      Gibson, Miss. Dorothy Winifred           female  22.0      0
409      Peacock, Miss. Treasteall                female   3.0      1
412      Henriksson, Miss. Jenny Lovisa           female  28.0      0

```

```

Parch      Ticket      Fare  Cabin  Embarked
4      1      3101298  12.2875  NaN      S
8      0      2657      7.2292  NaN      C
12     0      21228  82.2667  B45      S
15     0      SC/PARIS 2167  27.7208  NaN      C
18     0  STON/O2. 3101270  7.9250  NaN      S
..     ...      ...      ...      ...
385     2      220845  65.0000  NaN      S
395     0      13695  60.0000  C31      S

```

402	1		112378	59.4000	NaN	C
409	1	SOTON/O.Q.	3101315	13.7750	NaN	S
412	0		347086	7.7750	NaN	S

[71 rows x 12 columns]

```
[131]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] > 30) &
↳ (TitanDF['Sex'] == 'female')]
```

```
[131]:
```

	PassengerId	Survived	Pclass	\
1	893	1	3	
14	906	1	1	
19	911	1	3	
24	916	1	1	
32	924	1	3	
44	936	1	1	
48	940	1	1	
49	941	1	3	
59	951	1	1	
69	961	1	1	
74	966	1	1	
77	969	1	1	
96	988	1	1	
100	992	1	1	
112	1004	1	1	
114	1006	1	1	
122	1014	1	1	
141	1033	1	1	
153	1045	1	3	
178	1070	1	2	
179	1071	1	1	
206	1098	1	3	
208	1100	1	1	
213	1105	1	2	
214	1106	1	3	
218	1110	1	1	
224	1116	1	1	
239	1131	1	1	
240	1132	1	1	
241	1133	1	2	
305	1197	1	1	
309	1201	1	3	
313	1205	1	3	
314	1206	1	1	
324	1216	1	1	
330	1222	1	2	
343	1235	1	1	

347	1239	1	3
349	1241	1	2
350	1242	1	1
356	1248	1	1
362	1254	1	2
368	1260	1	1
371	1263	1	1
374	1266	1	1
375	1267	1	1
391	1283	1	1
397	1289	1	1
411	1303	1	1
414	1306	1	1

	Name	Sex	Age	SibSp	\
1	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	
14	Chaffee, Mrs. Herbert Fuller (Carrie Constance...	female	47.0	1	
19	Assaf Khalil, Mrs. Mariana (Miriam)"	female	45.0	0	
24	Ryerson, Mrs. Arthur Larned (Emily Maria Borie)	female	48.0	1	
32	Dean, Mrs. Bertram (Eva Georgetta Light)	female	33.0	1	
44	Kimball, Mrs. Edwin Nelson Jr (Gertrude Parsons)	female	45.0	1	
48	Bucknell, Mrs. William Robert (Emma Eliza Ward)	female	60.0	0	
49	Coutts, Mrs. William (Winnie Minnie" Treanor)"	female	36.0	0	
59	Chaudanson, Miss. Victorine	female	36.0	0	
69	Fortune, Mrs. Mark (Mary McDougald)	female	60.0	1	
74	Geiger, Miss. Amalie	female	35.0	0	
77	Cornell, Mrs. Robert Clifford (Malvina Helen L...	female	55.0	2	
96	Cavendish, Mrs. Tyrell William (Julia Florence...	female	76.0	1	
100	Stengel, Mrs. Charles Emil Henry (Annie May Mo...	female	43.0	1	
112	Evans, Miss. Edith Corse	female	36.0	0	
114	Straus, Mrs. Isidor (Rosalie Ida Blun)	female	63.0	1	
122	Schabert, Mrs. Paul (Emma Mock)	female	35.0	1	
141	Daniels, Miss. Sarah	female	33.0	0	
153	Klasen, Mrs. (Hulda Kristina Eugenia Lofqvist)	female	36.0	0	
178	Becker, Mrs. Allen Oliver (Nellie E Baumgardner)	female	36.0	0	
179	Compton, Mrs. Alexander Taylor (Mary Eliza Ing...	female	64.0	0	
206	McGowan, Miss. Katherine	female	35.0	0	
208	Rosenbaum, Miss. Edith Louise	female	33.0	0	
213	Howard, Mrs. Benjamin (Ellen Truelove Arman)	female	60.0	1	
214	Andersson, Miss. Ida Augusta Margareta	female	38.0	4	
218	Widener, Mrs. George Dunton (Eleanor Elkins)	female	50.0	1	
224	Candee, Mrs. Edward (Helen Churchill Hungerford)	female	53.0	0	
239	Douglas, Mrs. Walter Donald (Mahala Dutton)	female	48.0	1	
240	Lindstrom, Mrs. Carl Johan (Sigrid Posse)	female	55.0	0	
241	Christy, Mrs. (Alice Frances)	female	45.0	0	
305	Crosby, Mrs. Edward Gifford (Catherine Elizabe...	female	64.0	1	
309	Hansen, Mrs. Claus Peter (Jennie L Howard)	female	45.0	1	

313		Carr, Miss. Jeannie	female	37.0	0
314		White, Mrs. John Stuart (Ella Holmes)	female	55.0	0
324		Kreuchen, Miss. Emilie	female	39.0	0
330		Davies, Mrs. John Morgan (Elizabeth Agnes Mary...	female	48.0	0
343		Cardeza, Mrs. James Warburton Martinez (Charlo...	female	58.0	0
347		Whabee, Mrs. George Joseph (Shawneene Abi-Saab)	female	38.0	0
349		Walcroft, Miss. Nellie	female	31.0	0
350		Greenfield, Mrs. Leo David (Blanche Strouse)	female	45.0	0
356		Brown, Mrs. John Murray (Caroline Lane Lamson)	female	59.0	2
362		Ware, Mrs. John James (Florence Louise Long)	female	31.0	0
368		Gibson, Mrs. Leonard (Pauline C Boeson)	female	45.0	0
371		Wilson, Miss. Helen Alice	female	31.0	0
374		Dodge, Mrs. Washington (Ruth Vidaver)	female	54.0	1
375		Bowen, Miss. Grace Scott	female	45.0	0
391		Lines, Mrs. Ernest H (Elizabeth Lindsey James)	female	51.0	0
397		Frolicher-Stehli, Mrs. Maxmillian (Margaretha ...	female	48.0	1
411		Minahan, Mrs. William Edward (Lillian E Thorpe)	female	37.0	1
414		Oliva y Ocana, Dona. Fermina	female	39.0	0

	Parch	Ticket	Fare	Cabin	Embarked
1	0	363272	7.0000	NaN	S
14	0	W.E.P. 5734	61.1750	E31	S
19	0	2696	7.2250	NaN	C
24	3	PC 17608	262.3750	B57 B59 B63 B66	C
32	2	C.A. 2315	20.5750	NaN	S
44	0	11753	52.5542	D19	S
48	0	11813	76.2917	D15	C
49	2	C.A. 37671	15.9000	NaN	S
59	0	PC 17608	262.3750	B61	C
69	4	19950	263.0000	C23 C25 C27	S
74	0	113503	211.5000	C130	C
77	0	11770	25.7000	C101	S
96	0	19877	78.8500	C46	S
100	0	11778	55.4417	C116	C
112	0	PC 17531	31.6792	A29	C
114	0	PC 17483	221.7792	C55 C57	S
122	0	13236	57.7500	C28	C
141	0	113781	151.5500	NaN	S
153	2	350405	12.1833	NaN	S
178	3	230136	39.0000	F4	S
179	2	PC 17756	83.1583	E45	C
206	0	9232	7.7500	NaN	Q
208	0	PC 17613	27.7208	A11	C
213	0	24065	26.0000	NaN	S
214	2	347091	7.7750	NaN	S
218	1	113503	211.5000	C80	C
224	0	PC 17606	27.4458	NaN	C

239	0	PC	17761	106.4250		C86	C
240	0		112377	27.7208		NaN	C
241	2		237789	30.0000		NaN	S
305	1		112901	26.5500		B26	S
309	0		350026	14.1083		NaN	S
313	0		368364	7.7500		NaN	Q
314	0	PC	17760	135.6333		C32	C
324	0		24160	211.3375		NaN	S
330	2	C.A.	33112	36.7500		NaN	S
343	1	PC	17755	512.3292	B51 B53 B55		C
347	0		2688	7.2292		NaN	C
349	0	F.C.C.	13528	21.0000		NaN	S
350	1	PC	17759	63.3583	D10 D12		C
356	0		11769	51.4792	C101		S
362	0	CA	31352	21.0000		NaN	S
368	1		112378	59.4000		NaN	C
371	0		16966	134.5000	E39 E41		C
374	1		33638	81.8583	A34		S
375	0	PC	17608	262.3750		NaN	C
391	1	PC	17592	39.4000		D28	S
397	1		13567	79.2000		B41	C
411	0		19928	90.0000		C78	Q
414	0	PC	17758	108.9000		C105	C

```
[132]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] == 30) &
↳(TitanDF['Sex'] == 'female')]
```

```
[132]: PassengerId  Survived  Pclass  \
6            898         1         3
43           935         1         2
220          1112         1         2
291          1183         1         3
359          1251         1         3
400          1292         1         1
```

	Name	Sex	Age	SibSp	\
6	Connolly, Miss. Kate	female	30.0	0	
43	Corbett, Mrs. Walter H (Irene Colvin)	female	30.0	0	
220	Duran y More, Miss. Florentina	female	30.0	1	
291	Daly, Miss. Margaret Marcella Maggie"	female	30.0	0	
359	Lindell, Mrs. Edvard Bengtsson (Elin Gerda Per...	female	30.0	1	
400	Bonnell, Miss. Caroline	female	30.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
6	0	330972	7.6292	NaN	Q
43	0	237249	13.0000	NaN	S
220	0	SC/PARIS 2148	13.8583	NaN	C

291	0	382650	6.9500	NaN	Q
359	0	349910	15.5500	NaN	S
400	0	36928	164.8667	C7	S

```
[125]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] < 30) &
↳(TitanDF['Sex'] == 'Male')]
```

```
[125]: Empty DataFrame
Columns: [PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket,
Fare, Cabin, Embarked]
Index: []
```

```
[126]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] > 30) &
↳(TitanDF['Sex'] == 'Male')]
```

```
[126]: Empty DataFrame
Columns: [PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket,
Fare, Cabin, Embarked]
Index: []
```

```
[127]: TitanDF.loc[(TitanDF['Survived'] == 1) & (TitanDF['Age'] == 30) &
↳(TitanDF['Sex'] == 'Male')]
```

```
[127]: Empty DataFrame
Columns: [PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket,
Fare, Cabin, Embarked]
Index: []
```

```
[133]: TitanDF['SibSp'].value_counts()
```

```
[133]: SibSp
0    283
1    110
2     14
3      4
4      4
8      2
5      1
Name: count, dtype: int64
```

```
[ ]:
```

```
[ ]:
```

```
[ ]:
```